ASHOK

Senior Test Analyst & QA Automation Specialist

10+ Years | Test Automation | DevOps | Performance Testing | Telecom Domain Expert

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PROFESSIONAL SUMMARY

Telecom Testing & Automation Domain Expert with **10+ years** of specialized expertise in telecommunications systems testing, automation framework development, DevOps practices, and performance engineering. Proven track record as **Automation Lead & Test Lead** architecting enterprise-scale test solutions across 2G/3G/4G/5G networks. Demonstrated expertise in designing comprehensive automation strategies, leading QA teams, implementing CI/CD pipelines, and delivering high-performance, mission-critical telecom applications. Specialized in test automation, automation framework architecture, test leadership, performance testing, load testing, and DevOps infrastructure for telecom domain.

Domain Expertise for Target Roles:

- **Telecom Testing:** 10+ years hands-on experience in 2G/3G/4G/5G network testing, OSS/BSS systems, billing validation, and subscriber lifecycle management
- **Test Automation & Automation Lead:** Architected (WAF) achieving 85% test automation coverage; led automation teams; designed reusable test frameworks
- **Test Lead & QA Leadership:** Spearheaded QA strategy for major releases; managed SIT (System Integration Testing) for complex telecom projects; mentored 15+ QA engineers
- **Performance & Load Testing:** Architected load testing suites supporting 10,000+ concurrent users; implemented K6-based performance frameworks; built automated KPI dashboards
- **DevOps Engineering:** Led DevOps transformation reducing deployment by 40%; architected Jenkins CI/CD pipelines; designed Terraform IaC; optimized AWS/Kubernetes infrastructure
- **Advanced Metrics:** 70+ critical bugs identified (95% accuracy); 40% deployment time reduction; 85% test automation coverage; 95% uptime SLA achievement

CORE TECHNICAL COMPETENCIES

Test Automation & QA

- **Frameworks:** Robot Framework, Selenium, Appium, Cypress, Playwright, Karate
- **Languages:** Python, JavaScript, Java, Bash, Linux scripting
- **Protocols:** DIAMETER, RADIUS (GY, RO), FIX (V4.4+), SIP, REST, SOAP
- **Tools:** Postman, Swagger, SoapUI, JIRA, QC (HP ALM), Bugzilla

DevOps & Infrastructure

- **Cloud:** AWS (EC2, S3, Lambda, RDS), Azure, Keycloak
- **Orchestration:** Kubernetes (K8s), Docker, Docker Compose
- **IaC & Automation:** Terraform, Ansible, CloudFormation
- **CI/CD:** Jenkins (Pipeline as Code), Git, branching & conflict resolution
- **Monitoring:** ELK Stack, Splunk, Prometheus, Grafana

Performance & Load Testing

- **Tools:** JMeter, Blaze Meter, K6, WebLoad
- **Expertise:** Stress, Endurance, Soak, Spike testing
- **Skills:** Test script development (JMX), workload modeling, bottleneck analysis
- **Analysis:** Root cause analysis, root cause analysis (RCA)

Telecom Domain Expertise

- **Networks:** 2G/3G/4G/IoT networks, MVNO solutions, OSS/BSS
- **Platforms:** Nokia, Motorola UPP, Amadocs C1
- **Systems:** Provisioning, Billing, Mediation, OCS/RESTGW
- **Nodes:** HLR/HSS, PGW, DRA, AAA

TARGET ROLE SPECIALIZATIONS

Telecom Testing Engineer

2G/3G/4G/5G network testing • OSS/BSS systems • Billing validation • Subscriber lifecycle • Network node testing

Test Automation Engineer

Framework design • Cypress • Robot Framework • Python • API automation • GUI automation • 85% coverage achieved

Automation Lead

Team leadership • Framework architecture • Strategy design • Mentoring (15+ engineers) • WAF design • XTAF framework

Test Lead

SIT management • QA strategy • Release planning • Stakeholder reporting • QC tracking • Risk-based testing

DevOps Engineer

Terraform • Kubernetes • Jenkins CI/CD • AWS • Git • 40% deployment reduction • Infrastructure-as-Code

Performance & Load Test Engineer

JMeter • K6 • Load testing • Stress testing • 10,000+ concurrent users • Performance dashboards • KPI reporting

PROFESSIONAL EXPERIENCE

SENIOR TEST ANALYST

Capgemini UK PLC, Reading, UK

Enterprise-scale IoT platform enabling MVNOs to launch global managed 3G/4G services with integrated BSS (CMP) core network solutions supporting millions of enterprise subscribers

Automation & Framework Development

- **Full-Stack Automation Framework:** Designed and implemented comprehensive automation framework using Cypress for both API and GUI automation, integrating K6 for frontend and backend performance testing
- **Multi-Protocol Automation Suite:** Built Python + Robot Framework based automation suite covering GUI, API, and tool-based automation (Seagull, Spirent) to simulate Diameter and Radius protocol traffic for telecom systems
- **Network Node Automation:** Developed end-to-end automation framework for 4G, 5G, and IMS network nodes, enabling scalable, reusable, and CI/CD-ready test execution
- **4G Core Network Nodes:** Automated comprehensive 4G core network testing including PGW, SGW, MME, HLR with full functional and performance validation
- **5G Core Network Nodes:** Architected automation for advanced 5G infrastructure nodes including AMF, SMF, UPF, AUSF, UDM, NRF, PCF with real-time protocol validation
- **IMS Node Automation:** Implemented automated testing for IMS infrastructure nodes including ASBG, CSCF, vBGF with SIP protocol validation
- **Protocol-Level Automation:** Developed automation libraries for validating 4G, 5G, and IMS protocols including Diameter, Radius, SIP, GTP-C, GTP-U, HTTP/2, NGAP, PFCP, and telecom signaling protocols
- **Protocol Simulator Integration:** Integrated protocol simulators (Seagull, Spirent) into CI pipelines enabling high-volume signaling traffic simulation and validation
- **Performance & Reliability Engineering:** Implemented K6-based performance test suites for API, backend microservices, and UI flows with automated dashboards and reporting
- **Performance Metrics & KPIs:** Built comprehensive dashboards tracking latency, throughput, and load-based KPIs with automated reporting and SLA validation
- Architected **Automation Framework (WAF)** supporting GUI, API, and mobile automation achieving **85% test automation coverage** reducing manual testing by 70%
- Designed **XTAF (eXtendable Test Automation Framework)** with advanced scripting capabilities enabling rapid test case development and framework extensibility

DevOps & Infrastructure

- Designed comprehensive Terraform scripts defining infrastructure-as-code across cloud environments
- Proficient with AWS services (EC2, RDS, S3, Lambda) optimizing resource utilization and costs
- Authored complex Ansible playbooks automating application provisioning and deployment workflows
- Implemented Git workflows with branch strategies, reducing merge conflicts by 60%
- Built Jenkins Pipeline-as-Code architectures enabling continuous integration and delivery

API & Web Services Testing

- Executed extensive REST and SOAP web service testing using JMeter, SoapUI, Postman, Swagger
- Designed black-box and white-box test cases with comprehensive boundary value analysis
- Conducted functional and non-functional testing (performance, load, security) on critical APIs
- Managed API test automation in Robot Framework with Python, JSON, and REST clients

Performance & Load Testing Excellence

- Designed sophisticated load test scripts for REST and SOAP APIs measuring performance under variable workload conditions
- Built customized performance testing framework using JMeter, Blaze Meter, Jenkins, Tomcat, and Python
- Executed comprehensive load testing strategies: stress, endurance, soak, and spike testing
- Utilized ELK Stack and Splunk for real-time monitoring and performance analytics
- Identified and resolved system bottlenecks enabling **95% uptime SLA** achievement

Team Leadership & QA Strategy

- Spearheaded planning and execution of System DevOps, IaC, CI/CD, SIT, and performance testing for major releases
- Led cross-functional teams ensuring compliance with quality standards and STLC methodologies
- Expert in Agile, V&V, and Waterfall testing environments; skilled in risk-based and exploratory testing

SENIOR CONSULTANT I

System Integration Testing Lead for major releases managing functional, API, performance, and integration testing domains

Leadership & Testing Excellence

- Assumed role as **SIT Lead** managing comprehensive System Integration Testing for major releases
- Orchestrated end-to-end testing strategy covering functional, API, performance, and integration testing domains
- Established and maintained QA best practices, ensuring consistent quality across release cycles

Automation Framework Development

- Led development of robot framework automation suite integrating Python, Selenium, and Appium
- Designed mobile device provisioning and automation workflows reducing manual effort by 75%
- Implemented performance and load testing framework integrating JMeter, Webload, Blaze Meter

Subscriber Management & Billing Validation

- Managed end-to-end provisioning of mobile subscribers and offer application
- Executed Circuit-switched (CS) and Packet-switched (PS) call testing with billing validation
- Validated charging and mediation system accuracy ensuring financial integrity

Quality Metrics & Reporting

- Maintained rigorous QC tracking and timely issue reporting through JIRA and HP QC
- Conducted regular project reviews with internal stakeholders and customer technical teams
- Peer reviewed test cases ensuring comprehensive coverage and quality alignment

SENIOR TEST ENGINEER

IMS-based LTE standard Push-to-Talk system providing half-duplex communication over packet data networks with integrated HSS, CSCF, MRF capabilities

- Translated user requirements into comprehensive test cases and test beds
- Directed SIP-based call processing testing across feature, regression, and sanity test scenarios
- Executed positive/negative testing for iOS and Android mobile clients (UPP APP)
- Performed scratch ESXi installations on HP DL380 servers and system upgrades
- Managed HP VMware virtual machine environments
- Analyzed system logs (adb, tcpdump, application logs) for debugging and diagnostics
- Developed and executed Python automation scripts using Google Appium framework
- Tracked defects and issues using HP ALM and JIRA with comprehensive reporting

SENIOR TEST ENGINEER

Complex billing and charging solution supporting Offline Billing and Online Charging System (OCS) components

- Engineered comprehensive functional and integration test suites
- Validated GUI and system functionality across multiple build cycles
- Gained expertise in OSS/BSS billing systems: CSM, CRM, RC, NRC components
- Executed regression testing for each build cycle ensuring stability
- Analyzed application logs identifying root causes of failures
- Tracked and reported defects using IBM Clear-Quest
- Conducted unit testing and system testing activities

Capgemini, Bangalore, India | OSS/BSS Products & Infrastructure

- Progressed through roles gaining comprehensive QA foundation across diverse technologies
- Supported lab auditing and system observations on production environments
- Executed GUI and functional testing on OSS/BSS applications (Mediation, Web Services)
- Performed regression testing across multiple builds and versions
- Conducted web services testing using SoapUI with WSDL knowledge
- Managed scratch installations, system upgrades, and RPM package installations
- Executed positive/negative testing with comprehensive defect reporting (Bugzilla)
- Analyzed Linux and Java application logs for diagnostics and troubleshooting

CERTIFICATIONS & CONTINUOUS LEARNING

Professional Certifications & Training

- ✓ SAFe 5 Certified Practitioner – Scaled Agile Inc.
- ✓ Performance Testing with JMeter – Certified (Coursera)
- ✓ Python for Developers – Altran Developer Certification
- ✓ Advanced Automation Framework Development – Internal Training
- ✓ Kubernetes & Container Orchestration – AWS & Linux Academy
- ✓ DevOps & CI/CD Pipeline Architecture – Jenkins & Git Expertise

Education

Master of Computer Applications (MCA)
Bachelor of Computer Applications (BCA)
Higher Secondary (XII - Science)
Secondary School (SSLC - Science)

RECOGNITION & AWARDS

- **Individual Excellence Award (2020) – Recognized for valuable contributions to team success**
- **Team Excellence Award (2021, 2018, 2017) – Quality delivery and testing excellence**
- **Bug Raising Excellence (2017) – 70+ critical bugs identified with 95% accuracy rate**

LANGUAGE PROFICIENCY

Skill	English	Kannada	Telugu	Tamil	Hindi
Speak	Fluent	Native	Fluent	Fluent	Fluent
Read	Fluent	Native	Fluent	Fluent	Fluent
Write	Fluent	Native	Fluent	Fluent	Fluent

GLOBAL EXPERIENCE & EXPOSURE

Strategic Business Travel & Client Engagement:

- **EMEA Region: United Kingdom, Austria, France**
- **APAC Region: India, Singapore**
- **Multi-client Engagement: European, American, Latin American, and Asian enterprises**
- **Cross-functional Collaboration: Worked seamlessly with distributed global teams across multiple time zones**

ADDITIONAL STRENGTHS

✓ Problem Solving: Analytical mindset with strong debugging and troubleshooting capabilities

✓ Communication: Clear articulation of technical concepts to both technical and non-technical stakeholders

✓ Leadership: Mentor and guide QA engineers; foster collaborative team environments

✓ Adaptability: Thrive in dynamic Agile/Scrum environments with rapidly changing requirements

✓ Documentation: Create comprehensive technical specifications, test plans, and reports

✓ Attention to Detail: Meticulous approach ensuring minimal defects reach production

✓ Continuous Improvement: Proactive identification of process improvements and efficiency gains

References and additional project documentation available upon request
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